

Career Opportunities in Algorithms

اس باب کی مہارتیں سافٹ ویئر انجینئر، بیک اینڈ ڈیولپر، الگورتھم انجینئر، سسٹمز اینالسٹ اور کمپیٹیو پروگرامر کے...

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 24 of 24

Career Opportunities in Algorithms

LEARN IT

The skills in this chapter — proving correctness, writing clear modules, judging Big O, and choosing data structures — are the daily tools of several computing careers.

Career	What the person does
Software Engineer	Builds and maintains software that relies on data structures and efficient algorithms to solve logic problems.
Backend Developer	Builds server-side logic where choosing the right structure (array, queue) directly affects web performance.
Algorithm Engineer	Specialises in optimising algorithms for search engines, finance or logistics.
Systems Analyst	Evaluates and improves the efficiency and clarity of existing computer systems in an organisation.
Competitive Programmer	Solves complex algorithmic problems under time limits, a common path into top tech roles.

WORKED EXAMPLE

Worked example: match the task.

1. "Replace an $O(n^2)$ report with an $O(n \log n)$ sort so the payroll finishes overnight." → **Algorithm Engineer** (or a performance-minded Software Engineer).
2. "Put incoming HTTP requests into a FIFO buffer so the first visitor is served first." → **Backend Developer** (queue).
3. "Read a college's old fee software, find the nested-IF bottleneck, and recommend modules." → **Systems Analyst**.

REMEMBER IT

Every career in the table uses the same three tests you learned: **correct, clear, efficient** — plus the right data structure.

COMMON MISTAKE

Mistake: thinking only "Algorithm Engineer" needs Big O.

Fix: a backend developer who puts a nested loop on every page request will take a site down. Efficiency is a shared skill.

اردو میں سمجھیں

اس باب کی مہارتیں سافٹ ویئر انجینئر، بیک اینڈ ڈیولپر، الگورتھم انجینئر، سسٹمز اینالسٹ اور کمپیٹیو پروگرامر کے کام میں روز استعمال ہوتی ہیں: درست منطق، صاف ماڈیول، بگ او اور درست ڈیٹا سٹرکچر۔

CHECK YOURSELF

1. Which career specialises in search-engine or logistics algorithms?
2. Why does a backend developer care about queues and arrays?

Answers: 1. Algorithm Engineer. 2. The choice of structure (array, queue) directly affects web performance.

Next lecture: Chapter exam (in the chapter lecture PDF)